

HS-HU701 — Principles of Management

Semester VII • 2L:0T:0P • 2 Credits

Module 3 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1 Leadership	2
1.1 Concept and Nature of Leadership	2
1.2 Leadership Styles	2
1.3 Key Leadership Theories	2
1.3.1 Trait Theory	2
1.3.2 Behavioural Theories	3
1.3.3 Situational/Contingency Theories	3
2 Decision Making	4
2.1 Concept and Nature of Decision Making	4
2.2 Types of Decisions	4
2.3 The Decision-Making Process	4
2.4 Decision-Making Tools and Techniques	5
3 Economic, Financial and Quantitative Analysis	5
3.1 Production and Markets	6
3.2 National Income Accounting	6
3.3 Financial Function and Goals	6
3.4 Financial Statement Analysis	7
3.5 Quantitative Methods — Statistical Inference	8
3.6 Forecasting	8
3.7 Statistical Quality Control (SQC)	9

1. Leadership

Leadership — Definition

Leadership is the ability to influence, inspire, and guide others toward the achievement of organisational goals. A leader motivates followers to willingly exert effort toward common objectives.

Manager vs. Leader: A manager is appointed (formal authority); a leader may be formal or informal. Managers administer; leaders innovate. Managers maintain; leaders develop.

1.1 Concept and Nature of Leadership

- Leadership is a process, not just a position or trait.
- It involves a triad: **leader**, **followers**, and **situational context**.
- Effective leaders adapt their style to the readiness and needs of followers.
- Leadership is forward-looking; it involves creating a vision and aligning people toward it.

1.2 Leadership Styles

Style	Description	Best Suited For
Autocratic	Leader makes all decisions alone; high control, low participation.	Crises, unskilled workers, urgent tasks.
Democratic / Participative	Leader involves team in decision-making; shares authority.	Skilled/experienced teams; creative tasks.
Laissez-Faire / Free-Rein	Leader delegates fully; minimal intervention.	Highly skilled experts; research teams.
Transformational	Leader inspires change through vision, charisma, and individual attention.	Organisational change; long-term innovation.
Transactional	Leader motivates through reward and punishment; maintains the status quo.	Routine, well-defined tasks.
Servant Leadership	Leader prioritises the needs of followers; facilitates growth and well-being.	Service-oriented organisations, NGOs.

1.3 Key Leadership Theories

1.3.1 Trait Theory

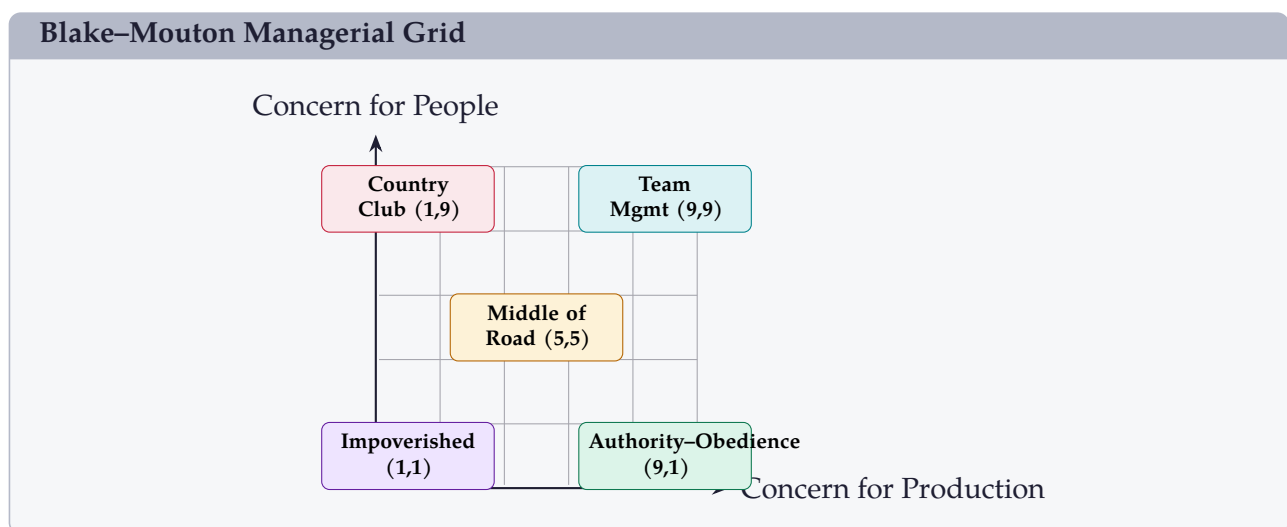
Leadership is determined by inherent personal qualities — physical, mental, and personality characteristics. Key traits identified: intelligence, self-confidence, integrity, decisiveness, drive, and sociability.

Limitation: Does not explain why people with the same traits succeed in some situations and fail in others.

1.3.2 Behavioural Theories

Focus on what leaders *do* rather than what they *are*:

- **Ohio State Studies:** Two dimensions — **Initiating Structure** (task-oriented) and **Consideration** (people-oriented). Best leaders are high on both.
- **Michigan Studies:** **Production-centred** vs. **Employee-centred** leadership. Employee-centred leaders achieve higher productivity and satisfaction.
- **Blake–Mouton Managerial Grid:** Plots leadership on two axes — concern for production (1–9) and concern for people (1–9). Ideal is (9,9) = Team Management.



1.3.3 Situational/Contingency Theories

Leadership effectiveness depends on the match between the leader's style and the situation:

- **Fiedler's Contingency Model:** Leadership effectiveness depends on the match between the leader's style (task vs. relationship oriented) and situational favourableness (leader–member relations, task structure, position power).
- **Hersey–Blanchard Situational Leadership:** Leader adapts style (Telling → Selling → Participating → Delegating) based on follower *readiness* (competence + commitment).
- **Path–Goal Theory (House):** Leader's job is to remove obstacles and provide guidance (directive, supportive, participative, achievement-oriented) to help followers reach their goals.

2. Decision Making

Decision Making — Definition

Decision Making is the process of identifying and selecting a course of action to solve a specific problem or capitalise on an opportunity. It is the core of managerial activity and involves choosing among alternatives under conditions of certainty, risk, or uncertainty.

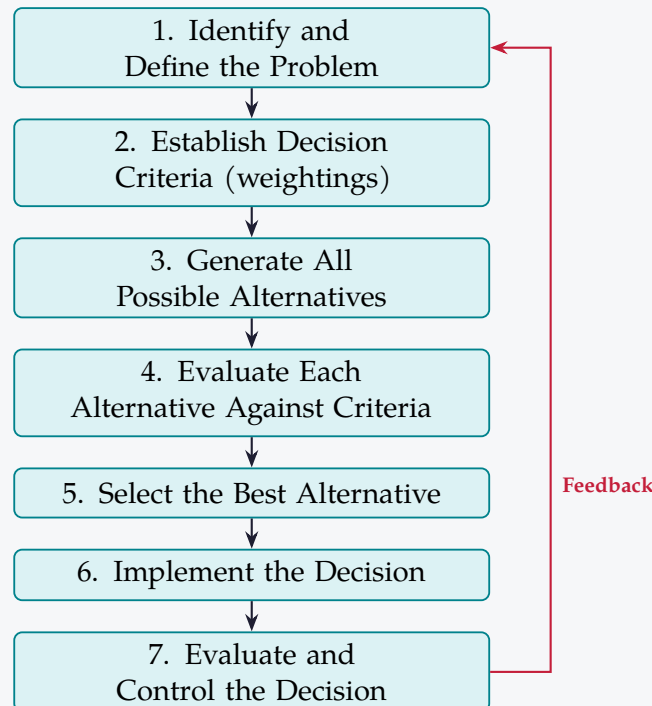
2.1 Concept and Nature of Decision Making

- **Pervasive:** Decision making occurs at all levels and in all functions of management.
- **Continuous:** Managers make decisions constantly throughout their working day.
- **Goal-Directed:** Every decision is aimed at solving a problem or achieving an objective.
- **Selection Among Alternatives:** Decision making implies the existence of two or more possible choices.

2.2 Types of Decisions

Type	Description	Example
Programmed (Structured)	Routine, repetitive decisions handled by standard procedures and rules.	Processing a standard purchase order.
Non-Programmed (Unstructured)	Novel, complex, non-routine decisions requiring judgment.	Entering a new international market.
Strategic	Long-term, top-level decisions affecting the entire organisation.	Mergers and acquisitions.
Tactical	Medium-term decisions by middle management to implement strategy.	Annual budget allocation.
Operational	Short-term, day-to-day decisions at the operational level.	Employee shift scheduling.

2.3 The Decision-Making Process

Rational Decision-Making Process**2.4 Decision-Making Tools and Techniques**

- **Brainstorming:** Group technique for generating maximum creative alternatives without criticism.
- **Delphi Technique:** Structured consensus-building among experts through successive anonymous questionnaires.
- **Decision Tree:** A graphical tool showing a sequence of decisions and their possible outcomes, payoffs, and probabilities.
- **Pay-off Matrix:** A table showing payoffs for each decision under each state of nature; used under uncertainty.
- **Cost-Benefit Analysis (CBA):** Compares the total expected costs and benefits of alternatives; choose the alternative with the highest net benefit.
- **Linear Programming (LP):** Optimises an objective function (maximise profit, minimise cost) subject to linear constraints.
- **SWOT Analysis:** Evaluates Strengths, Weaknesses, Opportunities, and Threats to guide strategic decisions.

3. Economic, Financial and Quantitative Analysis

3.1 Production and Markets

Production and Market Structures

Production is the process of transforming inputs (labour, capital, raw materials) into outputs (goods and services). Market structure defines the competitive environment in which firms operate.

Structure	Sellers	Price Control	Example
Perfect Competition	Many	None (price takers)	Agricultural markets.
Monopolistic Competition	Many	Some (product differentiation)	Restaurants, clothing.
Oligopoly	Few	Significant	Telecom, automobile.
Monopoly	One	Full (price maker)	Utility companies.

3.2 National Income Accounting

National Income Concepts

GDP (Expenditure) $GDP = C + I + G + (X - M)$

NNP at Market Price $NNP_{MP} = GNP_{MP} - \text{Depreciation}$

National Income (NI) $NI = NNP_{MP} - \text{Indirect Taxes} + \text{Subsidies}$

Personal Income $PI = NI - \text{Retained Earnings} - \text{Corp. Taxes} + \text{Transfer Payments}$

Disposable Income $DI = PI - \text{Personal Taxes}$

Where: C = Consumption, I = Investment, G = Government spending, $X - M$ = Net exports.

3.3 Financial Function and Goals

Financial Management & Functions

Financial Management deals with the procurement and efficient utilisation of funds to achieve the goals of an enterprise.

The three core **Financial Functions** (decision-making areas) are:

1. **Investment Decisions (Capital Budgeting):** Allocation of capital to long-term assets to generate future returns.
2. **Financing Decisions (Capital Structure):** Choosing the best mix of debt and equity to minimise the overall cost of capital.
3. **Dividend Decisions:** Deciding the proportion of net profits to distribute as dividends versus retaining them for reinvestment.

Goals of Financial Management:

The primary objective of financial management is guiding capital decisions. The two major approaches are:

- **Profit Maximisation:** Focuses on increasing net profits or earnings per share (EPS).
- **Wealth Maximisation (Shareholder Value Maximisation):** Focuses on increasing the net present value (NPV) of the firm's equity, reflected in the market value of its shares.

Aspect	Profit Maximisation	Wealth Maximisation
Primary Goal	Maximize accounting profits.	Maximize market price of shares (NPV of firm).
Time Value of Money	Ignores the timing of cash inflows.	Considers the timing of cash inflows (discounting).
Risk and Uncertainty	Ignores risk associated with future earnings.	Directly accounts for risk and uncertainty.
Focus	Short-term performance.	Long-term sustainable growth and value.
Limitation	Vague definition of "profit" (gross, net, pre-tax).	Requires efficient financial markets to reflect true value.

3.4 Financial Statement Analysis

Financial Statements

Balance Sheet: Snapshot of financial position at a point in time (Assets = Liabilities + Equity).

Income Statement (P&L): Shows revenues, expenses, and profit/loss over a period.

Cash Flow Statement: Tracks actual cash inflows and outflows from operations, investing, and financing activities.

Key Financial Ratios:

Liquidity	Current Ratio	Current Assets / Current Liabilities
Liquidity	Quick Ratio	(Current Assets – Inventory) / Current Liabilities
Profitability	Net Profit Margin	Net Profit / Net Sales × 100
Profitability	Return on Equity (ROE)	Net Profit / Shareholders' Equity × 100
Leverage	Debt-to-Equity	Total Debt / Shareholders' Equity
Activity	Inventory Turnover	Cost of Goods Sold / Average Inventory

3.5 Quantitative Methods — Statistical Inference

Statistical Inference

Statistical Inference is the process of drawing conclusions about population parameters based on data from a sample. Key concepts: hypothesis testing, confidence intervals, and significance levels.

Hypothesis Testing — Z-test for Population Mean

Test Statistic $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$

Standard Error $SE = \frac{\sigma}{\sqrt{n}}$

Confidence Interval $\mu \in (\bar{x} - z_{\alpha/2} \cdot SE, \bar{x} + z_{\alpha/2} \cdot SE)$

Reject H_0 if $|z| > z_{\alpha/2}$ (two-tailed test). Common: $z_{0.025} = 1.96$ for 95% confidence.

3.6 Forecasting

Forecasting

Forecasting is the process of using historical data, trends, and judgments to predict future values (demand, sales, costs). It reduces uncertainty in planning and decision-making.

Common forecasting methods:

- **Moving Average:** Average of the most recent n data points; smooths out fluctuations.
- **Exponential Smoothing:** Weighted moving average where more recent data receives greater weight.
- **Trend Projection:** Fits a least-squares regression line to historical data and projects it forward.
- **Qualitative methods:** Delphi, market surveys, jury of executive opinion — used when historical data is absent.

Simple Linear Regression (Trend Line)

Regression Line $\hat{y} = a + bx$

Slope $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$

Intercept $a = \frac{\sum y - b \sum x}{n}$

Correlation Coefficient $r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$

$|r| = 1$: perfect correlation; $r = 0$: no linear correlation. r^2 (coefficient of determination) measures the proportion of variance explained.

3.7 Statistical Quality Control (SQC)

Statistical Quality Control (SQC)

SQC is the application of statistical methods to monitor and control a production process so that the process operates at its full potential and produces conforming products.

Key tools of SQC:

- **Control Charts:** Monitor process variation over time. Central Line (CL), Upper Control Limit (UCL), and Lower Control Limit (LCL) are plotted. Common types: $\bar{X}$ -chart (mean), R-chart (range), p-chart (proportion defective).
- **Acceptance Sampling:** Inspecting a random sample from a batch; accept or reject the entire batch based on the sample result.
- **Process Capability Index (C_p, C_{pk}):** Measures how well a process meets specifications.

Control Chart Limits for $\bar{X}$ Chart

Centre Line (CL) $\bar{\bar{X}}$ (grand mean of all subgroup means)

Upper Control Limit $UCL = \bar{\bar{X}} + A_2 \bar{R}$

Lower Control Limit $LCL = \bar{\bar{X}} - A_2 \bar{R}$

A_2 = tabulated constant based on subgroup size n ; $\bar{R}$ = average range. A point outside UCL or LCL signals an **out-of-control** (assignable cause) condition.

Quick Revision — Module 3

Leadership, Decision Making & Quantitative Analysis | HS-HU701

Key Formulas at a Glance

GDP (Expenditure)	$C + I + G + (X - M)$
National Income	$GNP_{MP} - \text{Depreciation} - \text{Indirect Taxes} + \text{Subsidies}$
Z-test Statistic	$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$
Regression Slope	$b = (n \sum xy - \sum x \sum y) / (n \sum x^2 - (\sum x)^2)$
X-bar UCL / LCL	$\bar{\bar{X}} \pm A_2 \bar{R}$
Current Ratio	Current Assets / Current Liabilities
Net Profit Margin	Net Profit / Net Sales $\times 100$

One-Line Definitions

- **Leadership:** Ability to influence and guide others toward achieving goals.
- **Autocratic Leadership:** Leader makes all decisions unilaterally with full authority.
- **Transformational Leadership:** Inspires change through vision, charisma, and motivation.
- **Decision Making:** Process of selecting a course of action from available alternatives.
- **Programmed Decision:** Routine decision resolved by standard procedures.
- **Non-Programmed Decision:** Novel, complex decision requiring managerial judgment.
- **GDP:** Total monetary value of all goods and services produced within a country in a year.
- **SQC:** Use of statistical methods to monitor and control production quality.
- **Control Chart:** Graphical tool to detect process variation over time using UCL and LCL.
- **Regression Analysis:** Statistical method to model the relationship between variables.
- **Forecasting:** Using historical data and trends to predict future values.

Frequently Asked / Viva Questions

Q1. What is leadership? How does a leader differ from a manager?

Leadership is the ability to influence others toward goals. A manager has formal authority (appointed), while a leader may be informal. Managers administer; leaders innovate. Leaders inspire; managers control.

Frequently Asked / Viva Questions (contd.)**Q2. Name and explain the three basic leadership styles.**

Autocratic (leader decides alone), Democratic (team participates in decisions), Laissez-Faire (leader delegates fully with minimal intervention).

Q3. What is the Blake–Mouton Managerial Grid? What is the ideal position?

A grid plotting concern for production (x-axis) vs. concern for people (y-axis), each on a 1–9 scale. The ideal position is (9,9) Team Management — high concern for both people and production.

Q4. What is Hersey–Blanchard Situational Leadership theory?

Leaders adapt their style (Telling → Selling → Participating → Delegating) based on the follower's readiness level (combination of competence and commitment).

Q5. Distinguish between programmed and non-programmed decisions.

Programmed decisions are routine and repetitive, handled by standard rules (e.g., stock reorder). Non-programmed decisions are novel and complex, requiring judgment (e.g., entering a new market).

Q6. What is the Decision Tree technique?

A graphical representation of a decision problem showing decision nodes (squares), chance nodes (circles), possible outcomes, and associated probabilities and payoffs, enabling rational selection of the best path.

Q7. What is GDP? Write its expenditure-side formula.

GDP (Gross Domestic Product) is the total monetary value of all final goods and services produced within a country in a year. Expenditure formula: $GDP = C + I + G + (X - M)$.

Q8. What is Statistical Quality Control (SQC)? Name its key tools.

SQC is the use of statistical methods to monitor and control production processes. Key tools: Control Charts ($\bar{X}$ -chart, R-chart, p-chart), Acceptance Sampling, and Process Capability Indices.

Q9. What do UCL and LCL represent in a control chart?

UCL (Upper Control Limit) and LCL (Lower Control Limit) are the boundaries of natural process variation (set at $\pm 3\sigma$). A plotted point outside these limits indicates an out-of-control (assignable cause) condition.

Q10. State the formula for simple linear regression.

$\hat{y} = a + bx$, where b is the slope and a is the intercept. The slope is $b = (n \sum xy - \sum x \sum y) / (n \sum x^2 - (\sum x)^2)$.

Q11. What is the purpose of financial ratio analysis?

It allows comparison of a firm's performance against its own historical data, competitors, or industry benchmarks. Key categories: liquidity (current ratio), profitability (ROE), leverage (debt-to-equity), and activity (inventory turnover).

Frequently Asked / Viva Questions (contd.)**Q12. What is the difference between Transformational and Transactional leadership?**

Transformational leaders inspire change through vision and motivation (long-term focus). Transactional leaders use reward–punishment to maintain performance within existing norms (short-term, routine focus).

Types of Decisions — Comparison Table

Aspect	Programmed Decision	Non-Programmed Decision
Nature	Routine, repetitive	Novel, unique
Level	Lower/Middle management	Top management
Time	Short-term	Long-term
Basis	Standard rules, procedures	Judgment, creativity
Risk	Low	High
Example	Payroll processing	Product launch strategy